

Methodological issues in the development of a pharmacogenomic algorithm for warfarin dosing: comparison of two regression approaches

Aim: To ascertain whether multiple polynomial regression (MPR) has any advantage over multiple linear regression (MLR) in developing pharmacogenomic algorithms. **Materials & methods:** Two pharmacogenomic algorithms were developed based on MPR and MLR models from a warfarin pharmacogenomic data set (derivation cohort [n = 125] and validation cohort [n = 115]). **Results:** The MPR model showed better correlation with therapeutic dose ($r = 0.62$ vs 0.52); better diagnostic utility in distinguishing the warfarin-sensitive and warfarin-resistant patients (area under the receiver operating characteristic curves: 0.89 vs 0.81); and lower rate of underestimation (13.9 vs 20%) compared with the MLR model. Rate of overestimation was higher in the MPR than the MLR (10 vs 6.7%) model. **Conclusion:** The MPR approach has advantages over the MLR approach in predicting accurate and safe dose.

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Keywords: anticoagulation • multiple linear regression • multiple polynomial regression • pharmacogenomics • warfarin

Background

Warfarin is an oral anticoagulant that undergoes enantiomer-specific metabolism with CYP1A2 and CYP2C9 [1,2]. Warfarin exerts its anticoagulant effect by inhibiting the protein VKORC1 [3]. The efficacy of warfarin is affected primarily when metabolism of *S*-warfarin is altered [4]. Three SNPs, two in the *CYP2C9* gene (i.e., *CYP2C9**2 and *CYP2C9**3) and one in the *VKORC1* gene (i.e., -1639G>A) have been found to induce warfarin sensitivity [5–8]. The rare mutations in *VKORC1* were reported to be associated with clotting factor deficiencies and warfarin resistance [9]. The other crucial enzyme in this pathway is CYP4F2, which is a vitamin K1 oxidase. Carriers of the *CYP4F2* V433M allele have a reduced capacity to metabolize vitamin K1 (VK1) and hence require higher dose of warfarin [10]. Changes in thyroid hormone concentrations have the potential to significantly alter the extent of warfarin-induced anticoagulation [11].

Several pharmacogenomic algorithms were proposed to optimize warfarin dose. Most of the algorithms used the multiple linear regression (MLR) approach for the development of the pharmacogenomic algorithm. This is obviously owing to possibility of including up to ten predictor variables, and simplicity in computing [12–15]. However, apart from individual genetic effects on therapeutic dose of warfarin, there is likely to be complex gene–gene interactions between the variants that regulate pharmacokinetics and pharmacodynamics of warfarin [16]. Hence, we hypothesized that the multiple polynomial regression (MPR) approach might be superior to MLR approach in capturing synergistic effects of genetic variables. To test this hypothesis, we have developed two pharmacogenomic algorithms using MLR and MPR approaches along with the same data set. Furthermore, the International Warfarin Pharmacogenomic Consortium (IWPC) algorithm was also used for validation and comparison.

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Materials & methods

Patient enrolment

A total of 240 patients who were receiving warfarin were recruited for this study at Nizam's Institute of Medical Sciences, Hyderabad, India. Out of 240 patients, 125 patients (66 males and 59 females) who were receiving optimal dose of warfarin (international normalized ratio [INR]: 2.0–3.5) were enrolled into the derivation cohort and 115 (61 males and 54 females) patients who were not receiving optimal therapy (INR <2.0 and INR >3.5) were enrolled into the validation cohort. The mean ages of the derivation and validation cohorts were 38.4 ± 14.8 years versus 37.0 ± 15.5 years, respectively. Out of the 125 cases in the derivation cohort, 121 had valve replacement surgery, two had deep vein thrombosis and two had atrial fibrillation. All cases in the validation cohort had valve replacement surgery. The impact of standard empirical dosing on therapeutic index is shown in Figure 1. It depicts the number of patients receiving optimal therapy, lower dose and higher dose. The warfarin compliance-assessment scale was used to assess the compliance. The average compliance score of this cohort was 4 ± 2 . In this cohort, none of the subjects were using concomitant drugs that influence warfarin metabolism. Informed consent was obtained from all patients. The study protocol was approved by the institutional ethical committee of Nizam's Institute of Medical Sciences.

Clinical information, including age, gender, BMI, average daily maintenance dose of warfarin and information about dietary intake of vitamin K (e.g., turnip

greens, beetroot, broccoli, lettuce, cabbage, beef liver) were obtained from each patient. Based on the food frequency questionnaire, we calculated the dietary intake of vitamin K. To validate this assessment, vitamin K1, K2 and K3 levels were measured in plasma samples using reverse phase HPLC as described elsewhere [17]. Total vitamin K content in plasma correlated well with dietary vitamin K assessment ($r = 0.26$, $p < 0.0001$).

Genetic analysis of selected variants

Blood samples were collected in EDTA. Genomic DNA was isolated using phenol-chloroform extraction method. All the DNA samples were tested for *CYP2C9**2, *CYP2C9**3, *VKORC1**3, *VKORC1**4, *VKORC1* D36Y, *VKORC1* -1639G>A and *CYP4F2* V433M polymorphisms using PCR-RFLP approaches as described earlier [12,13]. The primers were designed to flank regions with only one known functional polymorphism at one time. RFLP experiments were designed to give more specific restriction digestion. Care was taken such that no other restriction sites were present in the amplicons for that particular restriction enzyme. Each PCR reaction was accompanied by a negative control without genomic DNA and a negative control (known as heterozygous). The quality of all the restriction enzymes was tested using plasmids with known restriction sites. Furthermore, genotyping was carried out by two independent researchers and observed 100% concordance. This analysis was carried out at Nizam's Institute of Medical Sciences.

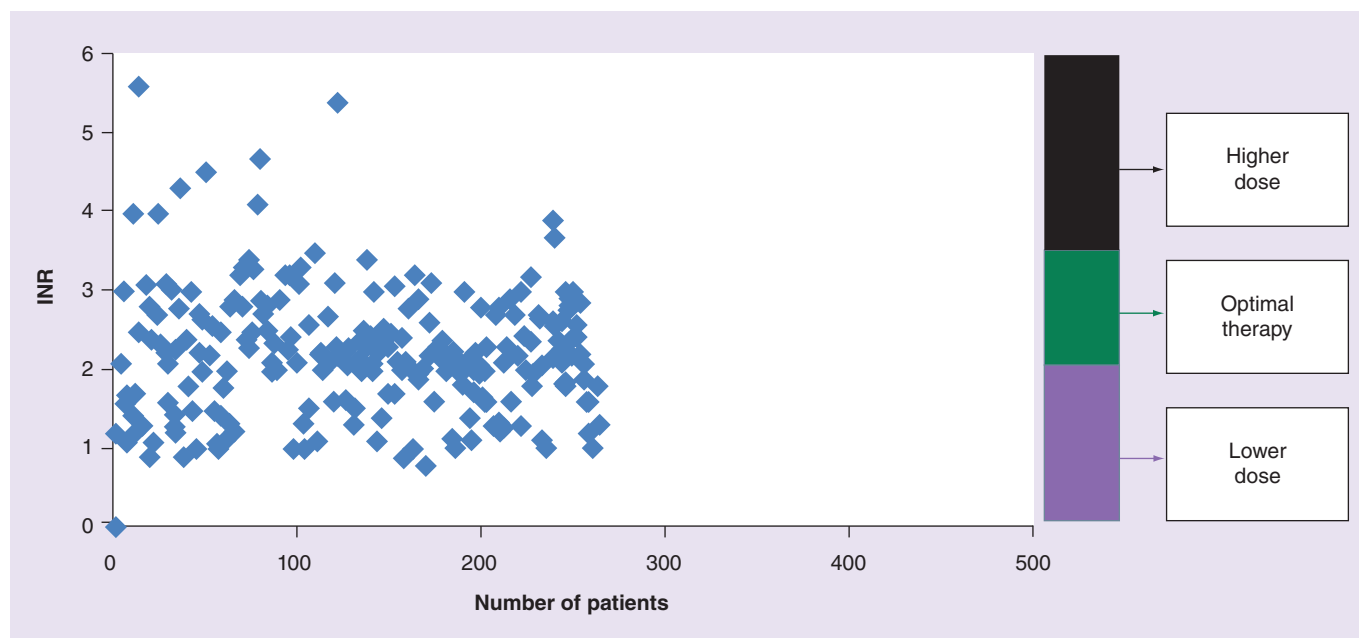


Figure 1. Effect of standard empirical dosing on therapeutic efficacy of warfarin. The x-axis represents the number of patients and the y-axis represents INR. INR values between 2.0–3.5 are indicative of optimal therapy with warfarin. INR >3.5 is indicative of higher dose of warfarin. INR <2.0 is indicative of lower dose of warfarin than required. INR: International normalized ratio.

Table 1. Univariate analysis: contribution of each variable towards therapeutic dose of warfarin.

Variable	Linear regression	Polynomial regression
Age (years)	$-0.12x + 37.97$	$-0.00414x^2 + 0.199x + 32.59$
BMI (kg/m ²)	$-0.40x + 36.29$	$-0.0433x^2 + 1.89x + 13.31$
Male gender	$4.94x + 30.47$	ND
CYP2C9*3	$-7.19x + 34.43$	$-1.56x^2 - 5.33x + 34.40$
CYP2C9*2	$6.55x + 32.11$	ND
VKORC1 -1639G>A	$-1.19x + 33.58$	$22.11x^2 - 26.24x + 34.01$
VKORC1*3	$-2.39x + 34.16$	$1.82x^2 - 4.58x + 34.23$
VKORC1*4	$0.96x + 33.08$	$10.42x^2 - 11.22x + 33.25$
CYP4F2 V433M	$0.166x + 33.11$	$-3.06x^2 + 6.21x + 31.58$
Vitamin K intake	$-8.03x + 35.46$	$-10.89x^2 + 1.35x + 34.50$
TSH (μIU/ml)	$-0.06x + 33.51$	$0.0832x^2 - 1.08x + 35.71$

ND: Not determined; TSH: Thyroid stimulating hormone; x: Variable.

Approaches used for development of pharmacogenomic algorithm

All variables (X1, X2,.....) were computed to study their impact on therapeutic dose (Y) using simple linear regression, multiple linear regression, MPR with the help of online computational website [18]. Simple linear regression was used for univariate analysis to evaluate the impact of each variable (X) on Y. This analysis helped in evaluating the positive and negative contribution of each variable towards therapeutic dose. We used the following criteria for inclusion of covariates in the regression models: the covariate has strong correlation with the Y ($p < 0.05$) in the univariate analysis; the collinearity was tested to decide which variable fits more in which model; we used stepwise regression for optimization of models; and we included few variables owing to their clinical relevance despite not being statistically significant predictors.

The MLR was used by considering 10 variables (demographic and genetic) as input variables and

therapeutic dose as a predictor variable. The predicted dose was correlated with the therapeutic dose. Three-step MPR was used for pharmacogenomic prediction of warfarin dose to ascertain whether these variables are in curvy-linear relationship with therapeutic dose.

Statistical analysis

We have compared the therapeutic and predicted dose of warfarin using Spearman's rank correlation coefficient (r). Bland–Altman analysis has been carried out for both MPR and MLR to know the bias and 95% CI limits of agreement between the therapeutic and predicted doses. Furthermore, it compares therapeutic dose with predicted dose using average versus difference plot. Receiver operating characteristics (ROC) curves were plotted for warfarin-sensitive versus warfarin-resistant predicted value and area under the curve (AUC) is used as index for clinical utility of these algorithms. All the statistical analyses were performed using GraphPad Prism (version 4.00 for Windows).

Box 1. Pharmacogenomic algorithms developed for predicting safe and effective doses of warfarin.

Multiple polynomial regression model

- $y = -0.00317 x_1^2 - 0.0155 x_1 x_2 + 0.0338 x_1 x_3 + 0.0278 x_1 x_4 - 0.0609 x_1 x_5 - 0.0851 x_1 x_6 - 0.0404 x_2^2 - 0.516 x_2 x_3 + 1.32 x_2 x_4 + 0.00540 x_2 x_5 + 0.555 x_2 x_6 - 1.05 x_3^2 + 0.891 x_3 x_4 + 18.19 x_3 x_5 - 2.96 x_3 x_6 + 4.60 x_4^2 - 7.13 x_4 x_5 - 1.13 x_4 x_6 + 13.86 x_5^2 + 13.86 x_5 x_6 - 4.60 x_6^2 + 0.528 x_1 + 1.54 x_2 + 3.82 x_3 - 35.52 x_4 - 23.81 x_5 - 0.738 x_6 + (3.51x_7) + (3.37x_8) + (0.87x_9) - (0.002x_{10}^2) - (0.31x_{10}) + 17.92$
- $x_1 = \text{age}; x_2 = \text{BMI}; x_3 = \text{CYP2C9*3}; x_4 = \text{VKORC1*3}; x_5 = \text{VKORC1*4}; x_6 = \text{CYP4F2}; x_7 = \text{gender}; x_8 = \text{CYP2C9*2}; x_9 = \text{VKORC1 -1639}; x_{10} = \text{TSH}; y = \text{therapeutic dose of warfarin in mg/week}$

Multiple linear regression model

- Therapeutic dose of warfarin (mg/week) = $-(0.13 \times \text{age}) + (0.0435 \times \text{BMI}) - (5.64 \times \text{CYP2C9*3}) + (0.0746 \times \text{VKORC1*3}) + (1.45 \times \text{VKORC1*4}) + (0.0325 \times \text{CYP4F2}) + (3.97 \times \text{male gender}) + (7.89 \times \text{CYP2C9*2}) - (3.49 \times \text{VKORC1 -1639}) - (8.37 \times \text{vitamin K intake}) - (0.31 \times \text{TSH}) + 38.27$

TSH: Thyroid stimulating hormone.

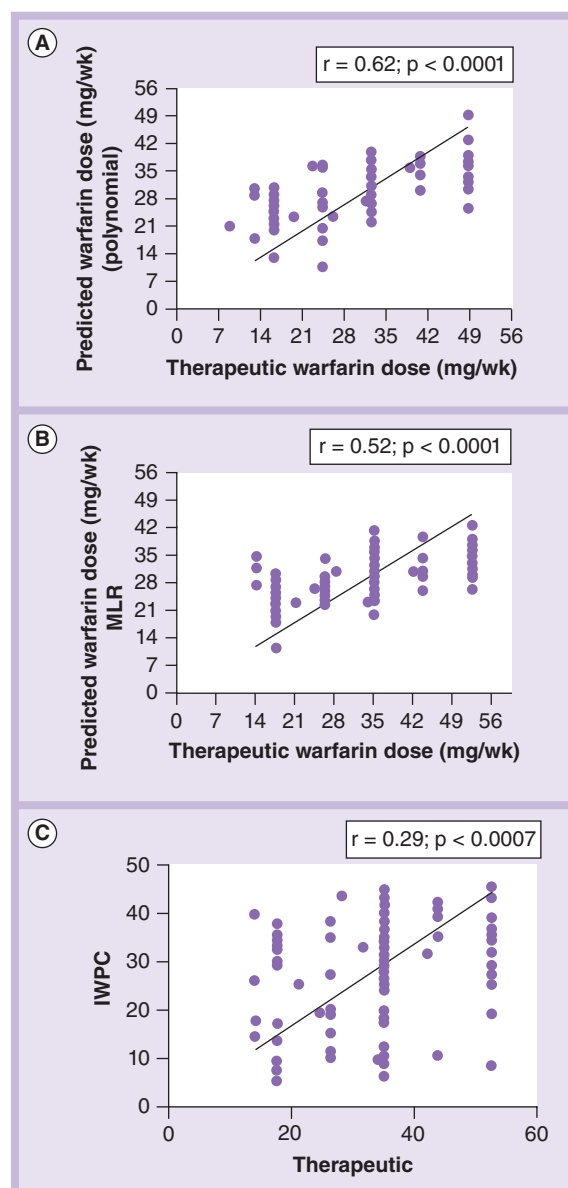


Figure 2. Correlation between therapeutic dose of warfarin versus predicted dose. Therapeutic dose of warfarin (on x-axis) was correlated with predicted dose (on y-axis) using three approaches: **(A)** multiple polynomial regression; **(B)** MLR; and **(C)** IWPC. r value is the Spearman's rank correlation coefficient. IWPC: International Warfarin Pharmacogenomic Consortium; MLR: Multiple linear regression.

Results

Univariate analysis

As shown in Table 1, both simple linear regression and polynomial regression models have showed the similar trend of association between each variable and therapeutic dose. The increase in age, increase in BMI, *CYP2C9**3, *VKORC1* -1639G>A and *VKORC1**3 genotype, vitamin K intake and increase in thyroid stimulating hormone require lower doses of warfarin.

Male gender, *CYP2C9**1/*2, *VKORC1**4 and *CYP4F2* V433M genotype require higher doses of warfarin. The two variables (i.e., gender and *CYP2C9**2 genotype) showed only a linear relationship and hence we could not obtain the polynomial regression equations corresponding to these variables.

Development of pharmacogenomic algorithm

As shown in Box 1, MLR and MPR approaches resulted in two different equations to calculate the predicted dose of warfarin in mg/week. Using the MPR model, we were unable to compute two variables (i.e., gender and *CYP2C9**2 genotype) due to their linear relationship with the therapeutic dose and hence the partial linear regression approach was used to incorporate them in the final model.

Correlation with therapeutic dose

As shown in Figure 2, in the derivation cohort, we have compared therapeutic and predicted dose using the MPR ($r = 0.62$; $p < 0.0001$) and MLR methods

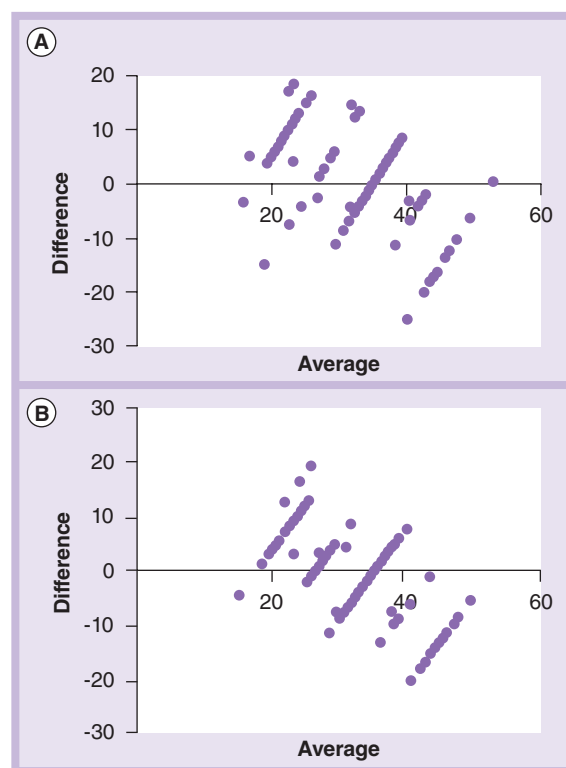


Figure 3. Bland and Altman analysis of therapeutic versus predicted dose. The x-axis represents the average of the therapeutic dose and predicted dose. The y-axis represents the differences between therapeutic and predicted dose. Two analyses were carried out: **(A)** corresponds to a multiple polynomial regression model; and **(B)** corresponds to a multiple linear regression model. The multiple polynomial regression model showed lesser differences between therapeutic and predicted doses.

($r = 0.52$; $p < 0.0001$). The MPR model showed better correlation than MLR, whereas the IWPC algorithm showed much less correlation ($r = 0.29$; $p < 0.0007$).

Bland–Altman analysis

As shown in Figure 3, Bland–Altman analysis was performed by plotting the average dose versus difference between the doses for therapeutic and predicted values. The MPR model showed lesser differences with therapeutic dose as compared with the MLR model.

ROC curves to test clinical utility of algorithms

As shown in Figure 4, the ROC curves were plotted to ascertain the discriminatory potential of newly developed algorithms in identifying warfarin-sensitive and warfarin-resistant patients. AUC values for MPR and MLR models were 0.87 and 0.81, respectively. The IWPC algorithm had lesser AUC (0.71) compared with the MPR and MLR models in distinguishing the warfarin-sensitive and warfarin-resistant patients.

Validation

In patients with $\text{INR} < 2.0$, the MPR model performed well with lesser rate of underestimation (13.9%) compared with the MLR (20%) and IWPC (23%) models. In patients with $\text{INR} > 3.5$, the IWPC model performed well by depicting the need of lower doses of warfarin in 96.7% of patients, while MPR and MLR models depicted the need of lowering the dose in 90 and 93.3% of patients, respectively.

Discussion

The results of the present study indicate that univariate analysis showed that younger subjects require higher doses of warfarin compared with older subjects. This is probably due to high metabolic rate during younger age. The male gender showed the association with high requirement for warfarin compared with female gender. With an increase in BMI, the amount of warfarin required was found to be decreased probably due to change in the warfarin uptake [19]. Furthermore, we have measured plasma warfarin and 7-hydroxywarfarin levels using HPLC across the different BMI groups. We observed that warfarin:7-hydroxywarfarin ratio increases with BMI ($r = 0.14$, $p = 0.03$) [PAVANI A, NAUSHAD SM, MANEMPATI AR, KUTALA VK, UNPUBLISHED DATA]. This suggests that with an increase in BMI, there is a decrease in warfarin metabolism. Consistent with the existing literature, *CYP2C9*3* and *VKORC1*-1639G>A polymorphisms showed association with warfarin sensitivity [12–15,20].

For warfarin dose prediction, we have used two approaches for the development of a pharmacogenomic algorithm (i.e., MPR and MLR models). To date, most

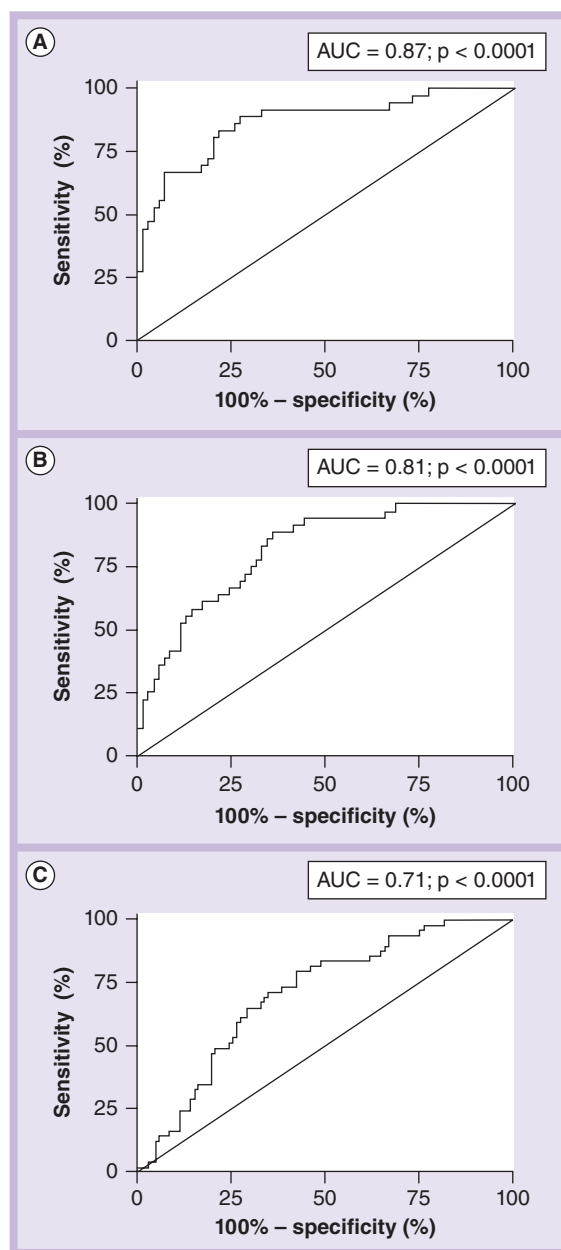


Figure 4. Receiver operating characteristic curves for warfarin sensitivity versus warfarin resistance. Using the standard dose of 4 mg/day as the cutoff, patients in the derivation cohort were segregated into two groups – that is, a warfarin-sensitive group (therapeutic dose of warfarin ≤ 4 mg/day) and a warfarin-resistant group (therapeutic dose of warfarin > 4 mg/day). Using the predicted doses of warfarin in these two groups, receiver operating characteristic (ROC) curves were plotted to ascertain diagnostic utility of pharmacogenomic algorithms in discriminating warfarin sensitivity versus warfarin resistance. (A) ROC curve of multiple polynomial regression model; (B) ROC curve of multiple linear regression model; and (C) ROC curve of International Warfarin Pharmacogenomic Consortium model. AUC served as index of diagnostic utility. AUC: Area under the curve.

studies have used MLR for warfarin prediction [12–15]. Although the predictor variables and outcome variables are the same in both approaches, we observed differences in the validation parameters.

In this study, we observed that the MPR model has better correlation compared with the MLR and IWPC models. In the IWPC derivation cohort, 1229 Asians were included [15]. However, there is no further information on the subgroups of Asians included in that study. Since IWPC was optimized to use across four racial groups (i.e. white, Asian, black or African–American groups) and missing or mixed racial groups [15], we included this model to evaluate whether it can be directly applied to Indians without the need of any other new algorithm. The very low ‘r’ value of the IWPC algorithm could be due to the different genetic make up of Indian subjects compared with the global population. The *VKORC1* D36Y polymorphism, which is known to be very frequent in the Jewish population [21,22], was found to be totally absent in our subjects [12,13] and rare in other populations [23]. We observed much lower frequencies of the *CYP2C9**2/*3 and *CYP2C9**3/*3 genotypes and none of the subjects had the *CYP2C9**2/*3 genotype.

The discriminatory potential of MPR was found to be greater than MLR and IWPC in identifying warfarin sensitivity versus warfarin resistance in these subjects. It could be due to the interaction between the different variables, which can be mathematically computed in only a polynomial model. In MLR, only the additive effects of each variable on therapeutic dose can be evaluated. The rate of underestimation was lower in MPR than MLR and IWPC and the rate of overestimation is higher in MPR than MLR and IWPC. The IWPC algorithm performed well in

predicting the need of lowering the dose of warfarin as it incorporated the data of many subjects who harbor genotypes that are associated with warfarin sensitivity.

The COAG (NCT00839657) and the EU-PACT (NCT01119300) randomized trials demonstrated mixed results of pharmacogenomic algorithms developed by the MLR approach in prolonging the time in the therapeutic range [24,25]. Results of the current study indicated that, our newly developed MPR approach provided better clinical utility compared with MLR analysis. In this context, we are conducting a randomized study in Indians to demonstrate the clinical utility of pharmacogenomic algorithms over the standard empirical dosing in prolonging the time in therapeutic range. In this work, we are also attempting to address whether the MPR approach has benefit and clinical utility over the MLR analysis in prolonging the time in therapeutic range.

Conclusion

Our study indicates that MPR and MLR are the two most popular techniques that can be used in the development of pharmacogenomic algorithms. They help in predicting effective dose and reducing adverse drug reactions. However, MPR has certain advantages in improving the accuracy and sensitivity of prediction and reducing rates of adverse drug reactions. MPR takes into account interactions between variables (i.e., gene–gene and gene–environmental interactions) thus facilitating precise prediction of warfarin dose.

Future perspective

The MPR model developed in this study can be utilized to develop pharmacogenomic algorithms with

Executive summary

Background

- Warfarin is an oral anticoagulant with a narrow therapeutic range and broad interindividual variability in dose requirements. Overestimation of dose causes bleeding events and underestimation of dose results in thrombosis.
- The differences in two modeling approaches, that is, multiple polynomial regression (MPR) and multiple linear regression (MLR), in precise prediction of warfarin dose were ascertained.

Materials & methods

- Patients receiving optimal therapy of warfarin (n = 125) were enrolled in the derivation cohort and patients with inappropriate dose (n = 115) were enrolled in the validation cohort.

Results & conclusion

- MPR model showed better correlation with therapeutic dose than MLR.
- Bland and Altman analysis showed lesser differences between the therapeutic and predicted dose in MPR compared with MLR.
- Receiver operating characteristic curves for warfarin sensitivity versus resistant groups showed better area under the curve in MPR than MLR.
- The rate of underestimation is lower in MPR than MLR and rate of overestimation is higher in MPR than MLR.

global applicability by incorporating large scale pharmacogenomic data generated from the different ethnic groups and populations with ethnicity as one of the predictors. This may ensure safe and effective dose prediction of warfarin.

Financial & competing interests disclosure

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subject matter or materials discussed in the manuscript apart from those disclosed.

No writing assistance was utilized in the production of this manuscript.

Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval or have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

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